

GyroSwin: 5D Surrogates for Gyrokinetic Plasma Turbulence Simulations

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github.com/ml-jku/neural-gyrokinetics

Abstract

Nuclear fusion plays a pivotal role in the quest for reliable and sustainable energy production. A major roadblock to viable fusion power is understanding plasma turbulence, which significantly impairs plasma confinement, and is vital for next-generation reactor design. Plasma turbulence is governed by the nonlinear gyrokinetic equation, which evolves a 5D distribution function over time. Due to its high computational cost, reduced-order models are often employed in practice to approximate turbulent transport of energy. However, they omit nonlinear effects unique to the full 5D dynamics. To tackle this, we introduce GyroSwin, the first scalable 5D neural surrogate that can model 5D nonlinear gyrokinetic simulations, thereby capturing the physical phenomena neglected by reduced models, while providing accurate estimates of turbulent heat transport. GyroSwin (i) extends hierarchical Vision Transformers to 5D, (ii) introduces cross-attention and integration modules for latent 3D $\leftrightarrow$ 5D interactions between electrostatic potential fields and the distribution function, and (iii) performs channelwise mode separation inspired by nonlinear physics. We demonstrate that GyroSwin outperforms widely used reduced numerics on heat flux prediction, captures the turbulent energy cascade, and reduces the cost of fully resolved nonlinear gyrokinetics by three orders of magnitude while remaining physically verifiable. GyroSwin shows promising scaling laws, tested up to one billion parameters, paving the way for scalable neural surrogates for gyrokinetic simulations of plasma turbulence.

1 Introduction

Nuclear fusion promises sustainable energy by fusing hydrogen isotopes in an ionized plasma. The plasma must be confined in a magnetic field as it reaches hundreds of millions of degrees. During this process, turbulence usually arises due to microinstabilities, resulting in energy and particle transport towards the reactor walls. The design and control of plasma scenarios strictly require knowledge of turbulent transport, which can be obtained via nonlinear gyrokinetic simulations that evolve a 5D Partial Differential Equation (PDE) over time.

The computational cost of nonlinear gyrokinetic simulations is prohibitive. Therefore, QuasiLinear approaches (QL), such as QuaLiKiz (Bourdelle et al., 2015; Citrin et al., 2017) and TGLF (Staebler et al., 2007; Staebler & Kinsey, 2010), are commonly used to approximate turbulent transport. They

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approach is to statistically model ensembles of particles in the plasma using a distribution function. This approach is termed plasma kinetics and is desirable to analyse stability in a bulk of plasma.

Plasma kinetics models the time evolution of electrons and ions via the distribution function f based on 3D coordinates, their parallel and perpendicular velocities, and the angle around the field lines. Gyrokinetics (Krommes, 2012) is a reduced form of plasma kinetics that is more efficient by averaging out the angle around the field lines and only considering the guiding center of a particle. Local gyrokinetics focuses on perpendicular spatial scales comparable to the gyroradius and on frequencies much lower than the particle cyclotron frequency, the circular motion of charged particles around magnetic field lines due to the Lorentz force. Hence, the 5D gyrokinetic distribution function can be written as $f = f(k_x, k_y, s, v_{\parallel}, \mu)$, where k_x and k_y are spectral coordinates (for the spatial x and y), s is the toroidal coordinate along a field line, and $v_{\parallel}$ and μ represent parallel and perpendicular velocity components, respectively. The perturbed time-evolution of f , for each species (ions and electrons), is governed by

$$\underbrace{\frac{\partial f}{\partial t} + (v_{\parallel} \mathbf{b} + \mathbf{v}_D) \cdot \nabla f - \frac{\mu B}{m} \frac{\mathbf{B} \cdot \nabla B}{B^2} \frac{\partial f}{\partial v_{\parallel}}}_{\text{Linear}} + \underbrace{\mathbf{v}_{\chi} \cdot \nabla f}_{\text{Nonlinear}} = S, \quad (1)$$

where $v_{\parallel} \mathbf{b}$ is the motion along magnetic field lines, $\mathbf{b} = \mathbf{B}/B$ is the unit vector along the magnetic field $\mathbf{B}$ with magnitude B^2 , $\mathbf{v}_D$ the magnetic drift due to gradients and curvature in $\mathbf{B}$, and $\mathbf{v}_{\chi}$ describes drifts arising from the $\mathbf{E} \times \mathbf{B}$ force, a key driver of plasma turbulence. The nonlinear term models the interaction between the distribution function f and the electrostatic potential, $-\nabla \phi = \mathbf{E}$, which comes from the velocity space integral of f itself, and describes turbulent advection. The resulting nonlinear coupling constitutes the computationally most expensive term. Finally, S is the source term that represents collisions between particles. For a more detailed derivation of the gyrokinetic equation, see Section A.

Quantities of interest. In practice, the main quantity of interest is the radial transport of energy (heat) towards the reactor wall, which is crucial for reactor design. It can be obtained by integrals on the 5D distribution function as

$$Q = \int \mathbf{C} \int \mathbf{v}^2 \phi \mathbf{f} dv_{\parallel} d\mu \, dx dy ds, \phi = \mathbf{A} \int \mathbf{J}_0 \mathbf{f} dv_{\parallel} d\mu, \quad (2)$$

where $\mathbf{A}, \mathbf{C} \in \mathbb{R}^{x \times s \times y}$ comprise geometric and operating parameters, $\mathbf{v} \in \mathbb{R}^{v_{\parallel} \times \mu}$ denotes particle energy, and $\mathbf{J}_0$ denotes the zeroth-order Bessel function and $\phi \in \mathbb{R}^{x \times s \times y}$ is the electrostatic potential. In a tokamak, $\mathbf{B}$ largely points in the toroidal direction, k_x encodes the radial direction, and k_y is perpendicular to k_x (binormal $\approx$ poloidal) in Fourier space. Electrostatic fluctuations that drive turbulence occur mainly in the k_y direction, while k_x encodes the radial structural properties of turbulence, i.e. the size of emerging eddies.

Radial turbulent transport occurs only if the amplitude of the k_y modes is non-zero, as this results in $\mathbf{E} \times \mathbf{B}$ drift, and consequently in fully developed nonlinear turbulence. A turbulent simulation usually follows a certain pattern. In the linear phase, different modes in k_y start growing due to underlying micro-instability, resulting in an initial increase in heat flux. Afterwards, the simulations enter the nonlinear (saturated) regime where modes start interacting and the system converges into a statistically steady state. This state is controlled by zonal flows that shear and break up arising eddies, effectively dampening turbulence (Itoh et al., 2006).

To analyse turbulence, practitioners usually investigate various spectra along the k_y direction, as those provide insights into turbulent transport. Most importantly, $Q(k_y)$ provides insight into which modes dominate the turbulence energy:

$$Q(k_y) = \sum_{v_{\parallel}, \mu, s, k_x} Q(v_{\parallel}, \mu, s, k_x, k_y), \quad (3)$$

where $Q \in \mathbb{R}^{v_{\parallel} \times \mu \times s \times k_x \times k_y}$ is the flux field, which aggregates to the scalar flux Q in Equation (2). In addition $W(k_y) \in \mathbb{R}^{k_y}$ describes the intensity of turbulence along y , and zonal flows dampen

²We adopt uppercase notation for vector fields $\mathbf{E}$ and $\mathbf{B}$ to adhere with literature.